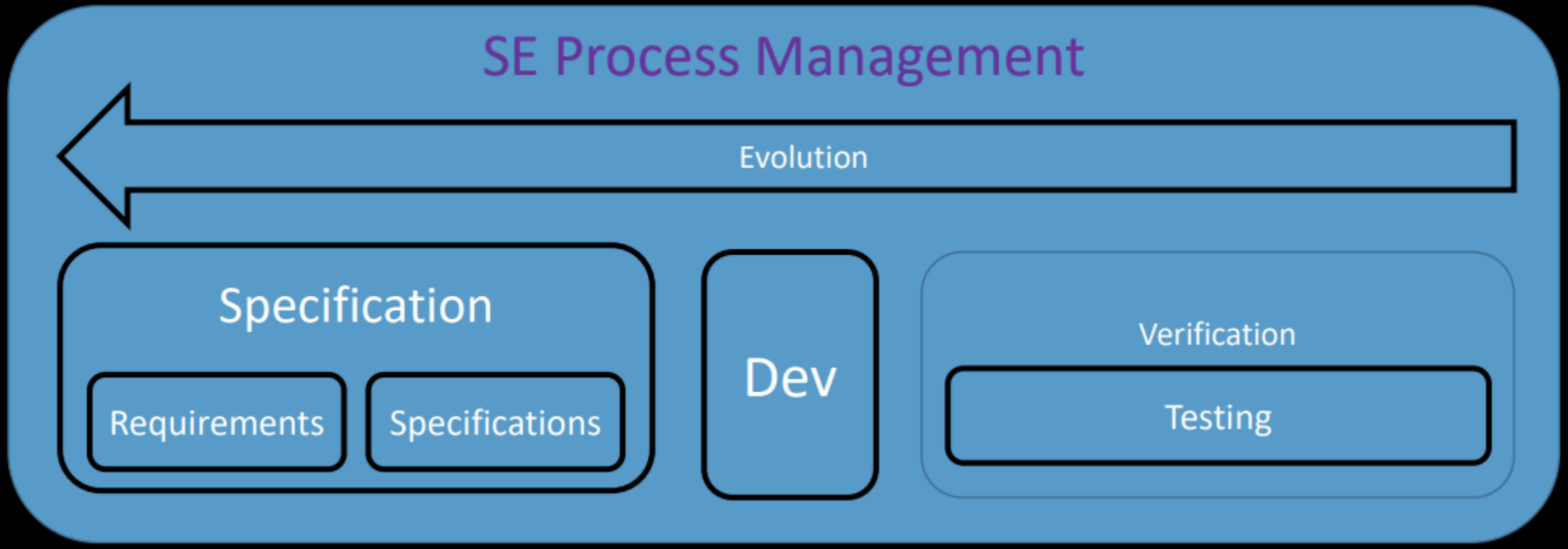
Software Engineering Management COMP4135

Seminar 9

Project Plan

Project Plan



Planning Diagrams

- PERT (1958) - tasks and dependencies
- Critical Path Method (1960) - risk analysis
- Gantt Chart (1910) - adding time to the tasks/dependencies
- Staff Allocation charts - who can do the tasks/dependencies

Practice

You are a project manager in a medium-sized software company. Your team has recently been assigned a project, and the company's senior management have asked you to coordinate it. After examining the details carefully, you have identified 16 major tasks that will need to be completed. You have determined the relationship among these tasks, in terms of which ones need to be completed before other tasks can begin. You have also estimated the amount of time(in days for a single engineer) that each task should require to complete. Your estimates are as follows:

1. Draw a PERT chart for these tasks, showing the task dependencies and durations, from the start to the end of the project (if appropriate, include a legend or explanation of there presentation).

2. Identifying a Critical Path for a project is important for a Project Manager.
   a. Identify and calculate the length of the Critical Path in this project.
   b. Explain why this Critical Path is important

Task	Duration (Days)	Dependencies
T01	8 days	
T02	15 days	T01
T03	10 days	T01
T04	5 days	T03
T05	10 days	T04
T06	15 days	T05
T07	5 days	T05 & T06
T08	5 days	T03
T09	5 days	T03 & T05
T10	1 day	T09
T11	5 days	T10
T12	4 days	T11
T13	5 days	T11
T14	5 days	T07 & T08 & T12 & T13
T15	2 days	T02 & T14
T16	1 day	T15

Practice

You are a project manager in a medium-sized software company. Your team has recently been assigned a project, and the company's senior management have asked you to coordinate it. After examining the details carefully, you have identified 16 major tasks that will need to be completed. You have determined the relationship among these tasks, in terms of which ones need to be completed before other tasks can begin. You have also estimated the amount of time(in days for a single engineer) that each task should require to complete. Your estimates are as follows:

3. Draw a Gantt Chart for the project – any risky areas? Presume you have a team of 8,working in (varying) pairs.

4. Create a staff allocation chart, indicating how staff time will be used over those times.

5. What is the impact of e.g. T06 overrunning to 20 days? Consider each chart you'veproduced as you ask this.a. Try this for a f tasks and find risky points.

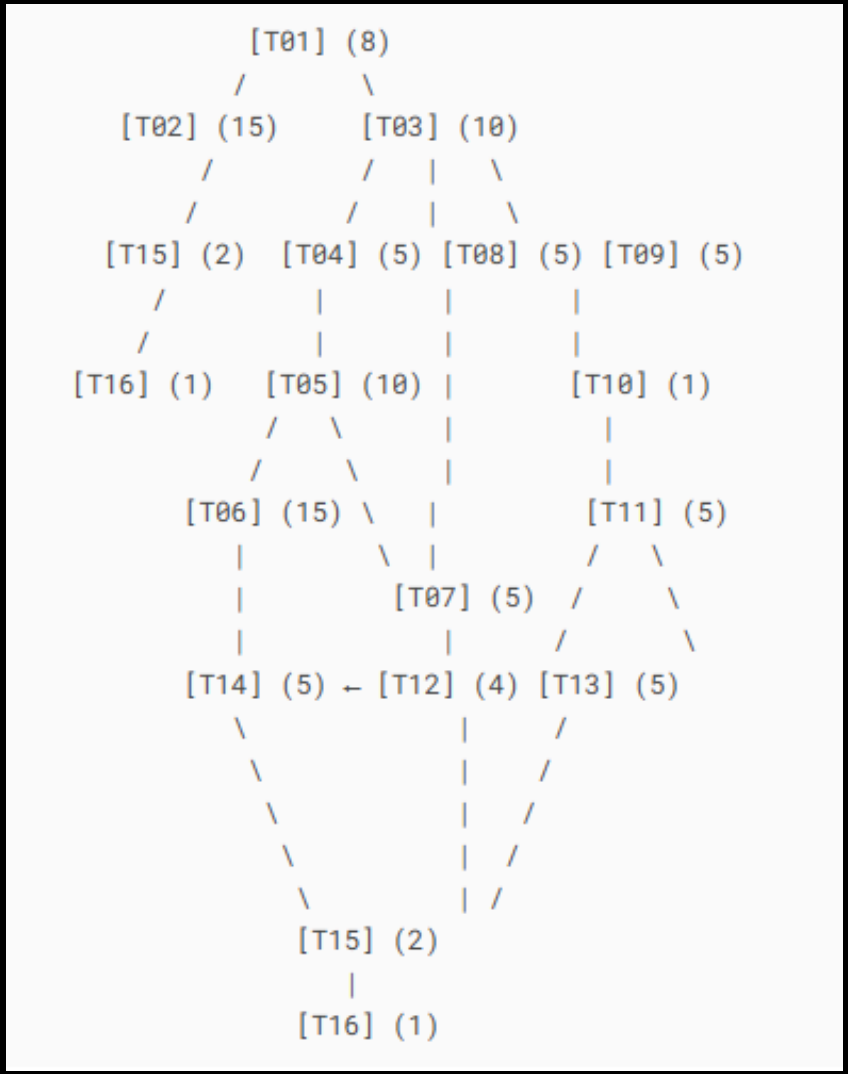
6. Price up the cost of project time based on ¥1200 per day for each person, considering people work on the project from the first point they are needed to the last point they are needed (when they might move to a different project).

7. If every task went 50% longer than expected – how long would the project be? What would the cost of that worse project be?

8. Multiply the cost of the project by 2.4 – to cover the overheads, what's the minimum or maximum costs now? Given that it should make a minimum 20% profit for the company,what should we charge for the project?

Practice

Task	Duration (Days)	Dependencies
T01	8 days	
T02	15 days	T01
T03	10 days	T01
T04	5 days	T03
T05	10 days	T04
T06	15 days	T05
T07	5 days	T05 & T06
T08	5 days	T03
T09	5 days	T03 & T05
T10	1 day	T09
T11	5 days	T10
T12	4 days	T11
T13	5 days	T11
T14	5 days	T07 & T08 & T12 & T13
T15	2 days	T02 & T14
T16	1 day	T15



Critical Path

[T01] → [T03] → [T05] → [T06] → [T07] → [T14] → [T15] → [T16]

61 days

Practice

Task	Duration	Days 1-10	Days 11-20	Days 21-30	Days 31-40	Days 41-50	Days 51-61	Notes
T01	8							Pair A
T02	15							Pair A (Critical)
T03	10							Pair B (Critical)
T04	5							Pair C
T05	10							Pair B (Critical)
T06	15							Pair C (Critical)
T07	5							Pair D (Critical)
T08	5							Pair D
T09	5							Pair A
T10	1							Pair A
T11	5							Pair B
T12	4							Pair C
T13	5							Pair D
T14	5							All Pairs (Critical)
T15	2							Pair A (Critical)
T16	1							Pair A (Critical)

Assessment !!! CW1

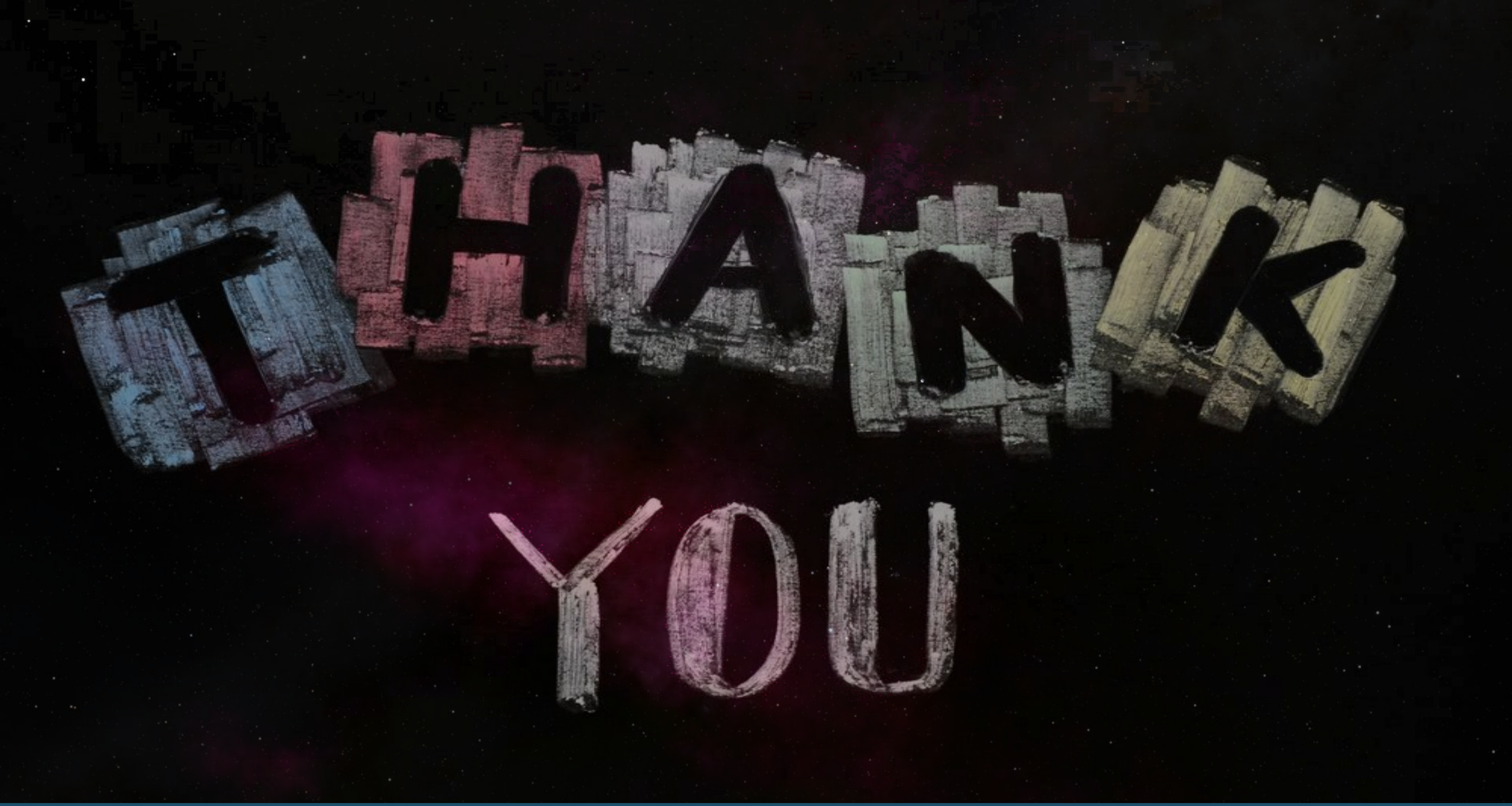
Group Task	Weight
Team Project Website	2%
Weekly Report	18%
Software	20%
Teamwork	10%
Final Group Report	15%
Video	10%
Presentation	15%

Reminder:

Submission Deadline: April 27th 2025, 12:00

- The website URL needs to be included in the final report.
- Teamwork is evaluated via GitHub/GitLab. Please add the module convenor and TA to GitHub/GitLab. GitHub homepage (readme.md) should display appropriate links, including the website, various reports, source code, documentation, etc.
- We will have group presentations next week. Please attend the lecture.
- All submissions related to the group's overall score should be submitted only once by the group leader.
- Peer assessment form needs to be submitted by each group member.

SET & SEM


THANK
YOU